

James Simon

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EDUCATION

University of Southern California

Expected May 2027

B.S. Computer Science & B.F.A. Game Design and Development

Dean's List 2023, 2024 | Academic Achievement Award 2024

Relevant Coursework: Artificial Intelligence, Deep Learning, AI for Sustainable Development (Fairness in ML), Algorithms, Computer Systems, Computational Physics (Bayesian Inference)

EXPERIENCE

OpenAI | AI Fellowship/Contract Work

09/2025–12/2025

- Collaborated on user intent recognition for image generation prompts enabling context-aware generation
- Designed scalable quality control frameworks ensuring dataset integrity for production AI systems
- Detected contaminated data batches maintaining model safety and preventing harmful outputs
- Led quantified QA testing cycles with unit tests ensuring consistent validation of production systems

INTERDEPENDENT Studios | NLP/ML Research Intern

03/2025–09/2025

- Authored 3 NLP/ML research papers under review focusing on alignment and model behavior
- Built multimodal ML framework supporting 2 multimillion-dollar film greenlighting decisions
- Developed NLP model predicting IMDb critic scores with 97% accuracy on 1,455 screenplays
- Improved prediction accuracy 24% over legacy version achieving 97% within ± 1 point
- Led six-person Scrum team conducting standups, sprint planning, and code reviews in Jira
- Automated pipeline processing 5,000+ screenplays using LLM APIs and webhook integrations

USC Rocket Propulsion Laboratory | Full Stack Team Lead

08/2024–Present

- Built mission control web app empowering world record-breaking student rocket launches
- Programmed embedded systems in C++ for sensor data parsing and control signal routing
- Implemented WebSocket connections for real-time telemetry streaming with Redis caching layer
- Automated testing pipelines with regression suites for mission-critical systems validation
- Migrated mission-critical web services during AWS outage ensuring system reliability

BEAD Lab | Research Assistant

Present

- Contributing to Study of Temperament and Attention Regulation examining attention bias in infants using EEG

RESEARCH & PUBLICATIONS

Uncovering Determinants of Obesity via Feature Importance | USC Ethical AI

Fall 2025

- Developed probability-based obesity risk prediction using CDC BRFSS dataset with 100,000+ responses
- Implemented dual modeling approach comparing classification vs regression with comprehensive metrics
- Performed comprehensive fairness evaluation addressing disparities across demographic groups
- Built risk stratification tool enabling healthcare providers to determine intervention timing

Preliminary Script Scorer (PSS) | USC Cinematic Arts & Viterbi

2024–2025

- Built ML model predicting critic reception using TF-IDF, Random Forest, Gradient Boosting on 1,455 scripts
- Improved accuracy 24% over legacy version achieving 97% predictions within ± 1 point
- Achieved 85% of predictions within ± 0.5 points through feature engineering and optimization

Funding vs Revenue Graph (FRG) | USC Research

2024–2025

- Trained neural network weighting model predictions by screenplay context improving accuracy 8% to $R^2=0.58$
- Curated dataset of 3,062 films for training revenue prediction models with confidence intervals

Fly Brain Connectome in Minecraft | USC Research Lab

Present

- Modeling exact neural connections of fly brain in Minecraft to achieve natural agentic behavior
- Implementing computer vision and reinforcement learning for behavioral adaptation

AWARDS & ACHIEVEMENTS

IBM Quantum Computing Hackathon Finalist | 2025

Dean's List | 2023, 2024 | Academic Achievement Award | 2024

TECHNICAL SKILLS

Programming & ML: Python, JavaScript, TypeScript, C++, Java, PyTorch, TensorFlow, scikit-learn, NLP, LLMs, Computer Vision, Generative AI, Reinforcement Learning

Systems & Tools: React, Flask, Node.js, SQL, Redis, Git, Docker, AWS, WebSockets, Embedded Systems, CI/CD, MLOps, Unity, Game Development, EEG, Quantum Computing (Qiskit)